

b.i. Describe the experimental verification of Davisson and Germer's diffraction experiment. 8 3 3 1,4

ii. What is the de-Broglie wavelength of a photon with a momentum 7.6×10^{-35} kgm/s? 2 4 3 1,4

29. a. Describe the Fraunhofer's diffraction at a single slit with neat diagram. 10 3 4 1,2

(OR)

b.i. Explain the production and detection of circularly polarized light using quarter wave plate? 6 4 4 1,2

ii. Calculate the plate thickness of a quarter wave plate for light of wave length 5.9×10^{-7} m. The refractive index of ordinary and extraordinary ray is 1.544 and 1.553 respectively. 4 4 4 1,3

30. a. Describe the principle, construction and working of CO₂ Laser with necessary diagrams. Explain the different mode of vibrations of CO₂ molecule. 10 3 5 1,3

(OR)

b. Derive an expression for Numerical aperture and acceptance angle for an optical fiber with necessary diagram. 10 4 5 1,3

Reg. No.

B.Tech. DEGREE EXAMINATION, JULY 2022

Frist, Second and Fourth Semester

18PYB101J – PHYSICS: ELECTROMAGNETIC THEORY, QUANTUM MECHANICS, WAVES AND OPTICS

(For the candidates admitted from the academic year 2018-2019 to 2019-2020)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B** should be answered in answer booklet.

Time: 2½ Hours

Max. Marks: 75

PART – A (25 × 1 = 25 Marks)

Answer **ALL** Questions

- | | Marks | BL | CO | PO |
|---|-------|----|----|-----|
| 1. The vector field whose curl is zero is called _____.
(A) Irrotational (B) Conservative
(C) Rotational (D) Solenoid | 1 | 1 | 1 | 1,2 |
| 2. Dielectrics are _____.
(A) Electric insulators (B) Electric conductors
(C) Materials that work under low voltage (D) Semimetals | 1 | 2 | 1 | 1,2 |
| 3. The Stoke's theorem uses which of the following operation _____.
(A) Divergence (B) Gradient
(C) Curl (D) Laplacian | 1 | 2 | 1 | 1,2 |
| 4. _____ is existence of permanent dipole in the absence of electric field.
(A) Non-polar electrics (B) Polar dielectrics
(C) Ionic dielectric (D) Polarizability | 1 | 1 | 1 | 1,2 |
| 5. The displacement current per unit area is called _____.
(A) Conduction current density (B) Displacement current density
(C) Permittivity (D) Permeability | 1 | 1 | 1 | 1,2 |
| 6. _____ magnetic material does not possess permanent magnetic moment.
(A) Para (B) Ferro
(C) Dia (D) Antiferro | 1 | 1 | 2 | 1,2 |
| 7. A tiny movable magnetized cylindrical volume in thin magnetic material is called _____.
(A) Magnetopplembites (B) Magnetic bubble
(C) Ferrites (D) Garnet | 1 | 1 | 2 | 1,2 |
| 8. Magnetopplembites belongs to a family of _____.
(A) Ferrites (B) Diamagnet
(C) Paramagnet (D) Conductor | 1 | 2 | 2 | 1,2 |

9. Which one of the following technique is mostly used to fabricate magnetic thin films? 1 1 2 1,2
(A) Cutting (B) Slicing
(C) Sputtering (D) Mechanical process
10. Garnets crystallize in _____ system. 1 2 2 1,2
(A) Isometric (B) Hexagonal
(C) Orthorhombic (D) Monoclinic
11. According to corpuscular theory light consists of tiny perfectly elastic particles called _____. 1 1 3 1,4
(A) Corpuscles (B) Photons
(C) Phonons (D) Quanta
12. _____ effect refers to the change in the wavelength of scattered X-rays by a material 1 2 3 1,4
(A) Photoelectric effect (B) Compton effect
(C) de-Broglie concept (D) Black body radiation
13. The energy level of an electron in 1D box are _____. 1 2 3 1,4
(A) Discrete (B) Continuous
(C) Random (D) Unified
14. A perfect black body is said to be a system which _____ all possible wavelengths of radiation. 1 1 3 1,4
(A) Transmits (B) Creates photon
(C) Absorbs and Emits (D) Diffracts
15. A variable quantity which characterizes de-Broglie wave is known as _____. 1 1 3 1,4
(A) Photon (B) Phonon
(C) Field (D) Wave function
16. In Fresnel's diffraction, the incident wave front should be _____. 1 2 4 1,2
(A) Elliptical (B) Plane
(C) Spherical (D) Cylindrical
17. The phenomenon of superposition of two coherent wave is called _____. 1 1 4 1,2
(A) Diffraction (B) Interference
(C) Reflection (D) Transmission
18. Nicol prism is made from _____ crystal 1 1 4 1,2
(A) Calcite (B) Nickel
(C) Cobalt (D) Zinc
19. Half wave plate produces a phase difference _____ between the waves. 1 2 4 1,2
(A) 45 degree (B) 90 degree
(C) 180 degree (D) 270 degree
20. Brewster's Law in terms of refractive index can be expressed by _____. 1 1 4 1,2
(A) $\mu = \sin ip$ (B) $\mu = \tan ip$

(C) $\mu = \cot ip$

(D) $\mu = \cos ip$

21. A pair of mirrors placed on either side of active medium is _____. 1 1 5 1,3
(A) Cavity (B) Optical resonator
(C) Population inversion (D) Active medium
22. The wave length emitted by Nd:YAG laser is _____. 1 2 5 1,3
(A) 10.6 μm (B) 1.06 μm
(C) 0.86 μm (D) 16.6 μm
23. The minimum population inversion density required to overcome the losses is called _____ population inversion. 1 1 5 1,3
(A) Threshold (B) Normal
(C) Standard (D) Dense
24. The principle used in optical fiber is _____. 1 1 5 1,3
(A) Polarization (B) Interference
(C) Diffraction (D) Total internal reflection
25. In step – index fibers, the signal travel in a _____. 1 2 5 1,3
(A) Zig-Zag manner (B) Random manner
(C) Skew manner (D) Linear manner

PART – B (5 × 10 = 50 Marks)

Answer ALL Questions

Marks BL CO PO

26. a. Obtain Maxwell's equation for electromagnetism from Gauss law of electrostatics and Ampere's circuital law. 10 4 1 1,2

(OR)

- b. Apply the concept of various polarization and derive the Langevin-Debye equation. 10 3 1 1,2

27. a.i. Explain the regular spinel and inverse spinel structures of Ferrites. 8 4 2 1,2

- ii. A magnetic field of 1800 amp/m produces a magnetic flux of 3×10^{-15} weber in an iron bar of cross section area 0.8 cm^2 . Calculate the permeability. 2 3 2 1,2

(OR)

- b.i. Explain the function of Giant magnetoresistance and Colossal magnetoresistance. 8 4 2 1,2

- ii. A magnetic field strength of 2×10^5 amperes / metre is applied to a paramagnetic material with a relative permeability, of 1.01. Calculate the value of magnetization. 2 3 2 1,2

28. a. Derive an expression for time – independent Schrodinger's wave equation. 10 4 3 1,4

(OR)